

FALCON's Response to RFP Flipkart for Supply of SWEDI + ARM Sorter System

June 2, 2023



www.falconautotech.com



**Kind Attention – Mr. Arvind Seshadri
Mr. Hemanth Kumar
Mr. Sandesh Mulik
Mr. Palvannan K**

M/s Flipkart Private Limited

Location – Anjaneya

Offer Ref: F23_00127; Date 02.06.2023

Subject – Techno-Commercial Offer for SWEDI+ARM Sorter

Dear Team,

We are pleased to submit our Techno-Commercial Offer in response to your RFP

As you will note, we have studied your requirements in great depth and, along with the information collected during the workshops and calls with you, we have put together a detailed technical proposal laid out in various sections and sequenced to enable you to understand our proposed solution and to re-enforce our commitment to be your partner in this strategic initiative.

In subsequent sections, we have highlighted the capabilities and experiences of Falcon Autotech with sections on our Automation Technologies and references.

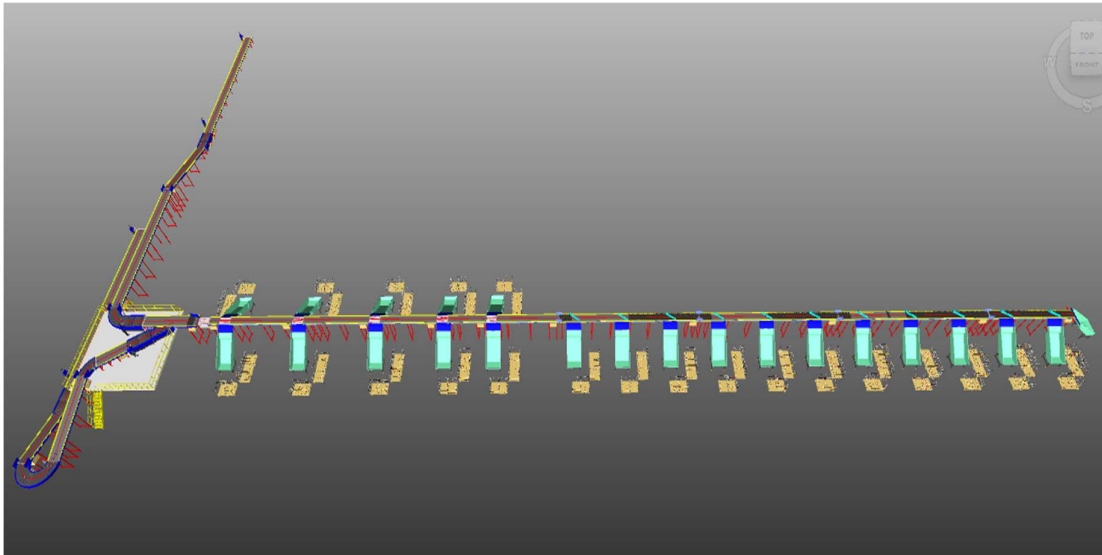
To conclude, I would like to add my personal commitment, on behalf of Falcon Autotech. As we move through the RFP process, please do not hesitate to contact me and my team. We will be pleased to assist you for any further information or clarifications that you might have.

Contact Person : Sanyog Pratap Singh
Title : Assistant Manager
Mobile : +91-8750052591
Email : Sanyog.singh@falconautoonline.com

Company Name : Falcon Autotech Pvt. Ltd.
Regd. Office : 57, Nimri Colony, Double Storey Flats, Delhi – 110052
Website : www.falconautotech.com



Response to RFP for SWEDI+ARM Sorter_ 1500 PPH



Proposal Reference – F23-00127

Falcon Autotech Private Limited
Plot No 308 & 309
Ecotech-1 Extension, Greater Noida
Uttar Pradesh- 201308
India.

Sanyog Pratap Singh
Asst. Manager
+91-8750052591
sanyog.singh@falconautoonline.com



Table of Contents

1. Glossary	6
2. About Falcon	7
3. Falcon's Experience and Achievements in Sortation Space Globally	19
4. Reference projects	20
4.1 Project 1- (Client – Courier Express Parcel, Dubai & Jeddah)	20
4.2 Project 2- (Client – E-Commerce, India)	21
4.3 Project 3- (CEP Client, Riyadh).....	22
4.4 Project 4- (CEP Client, UK).....	23
4.5 Project 5- (CEP Client, Sydney).....	24
4.6 Project 6- (Ecommerce Client, India)	25
4.7 Project 7- (E-commerce, India)	26
5. Handled Parcel Spectrumins Minimum Maximum	27
5.1 Product size Conveyable on the entire system.....	27
5.2 Parcels/Totes to be loaded on Conveyor/Sorter shall have the following characteristics:27	
5.3 Parcel not loadable on the system	28
5.4 Types of Products	28
6. Proposed System Description.....	29
6.1 Objective	29
6.2 Summary of the System	29
6.3 Main benefits of the proposed solution	29
7. Layout View.....	30
8. Proposed System Capacity	31
8.1 System Capacity	31
9. System description	31
9.1 Loading conveyor	31
9.2 Manual Scanning	32
9.3 Parcel Induction	32
9.4 Outbound Sorting	33
10. System Components	34
10.1 PVC Belt Conveyor	34
10.2 Plastic Modular Belt Conveyor	36
10.3 Swivel Wheel Sorting Unit - Falcon.....	37



10.4	ARM Sorting Unit	37
10.5	Powered Belt Merges & ZTR Curves (Pic for ref.)	38
10.6	Handheld Barcode Scanners	38
10.7	Collection Type Chute	39
10.8	Operator's Platform	39
10.9	Stair case	40
11.	Proposed System Technical Details/BOM	40
11.1	Mechanical Equipment	40
11.2	Electrical Equipment	41
11.3	Control System	41
12.	Falcon's WCS CONTROLIT	42
13.	High Availability Server Specifications (In Flipkart's Scope)	43
14.	Key Component Makes	44
15.	Safety	45
15.1	Push Buttons	45
15.2	Program Management	45
16.	Commercial Terms	46
16.1	Price Sheet	46
16.2	Payment Terms	46
16.3	Delivery timeline	47
17.	Warranty Period	47
18.	Exclusions	48



1. Glossary

S. No.	Term	Description
1	RFP	Request For Proposal
2	PPH	Parcels Per Hour
3	DBO	Damaged Barcode
4	MEZZ	Mezzanine
5	AC	Alternating Current
6	DC	Direct Current
7	PLC	Programmable Logic Controller
8	IT	Information Technology
9	BOQ	Bill Of Quantity
10	I/O	Input/ Output
11	PDP	Power Distribution Panel
12	PC	Personal Computer
13	UPS	Uninterrupted Power Supply
14	SWEDI	Swivel Wheel Divert



2. About Falcon

THE COMPANY

ABOUT US

Falcon Autotech is one of the leading Intra-logistics automation solution providers with a sharp focus on Parcel, Piece and Case Handling applications. We have a broad portfolio of Sortation Systems for Retail, E-Commerce and CEP (Courier, Express and Postal) industries and have supplied our systems across various countries globally.

OUR VISION

To create a global impact on the society by facilitating world's leading organizations run their supply chains efficiently by leveraging technology enabled products and solutions and, in the process, establish Falcon Autotech as one of the world's most respected and trusted brands in Intra-logistics warehouse automation space.

OUR JOURNEY

The team started out in 2004 solving special purpose automation problems for clients and later pivoted to building a standard technology stack spanning across Hardware, Firmware and Software to tackle bigger Supply Chain problems. Over the last few years, the organization has made rapid strides and has carved out a niche in some of the world's most cutting-edge technologies: Sortation, Robotics, Conveying, Vision Systems and IOT.

OUR CORE COMPETENCIES

Organizations are built and defined by the people they are able to attract, cultivate and retain and the talent management philosophy at Falcon Autotech could not agree more. At, Falcon Autotech we take a lot of pride in being the preferred career choice for some of the finest talent in the country across multiple domains such as:

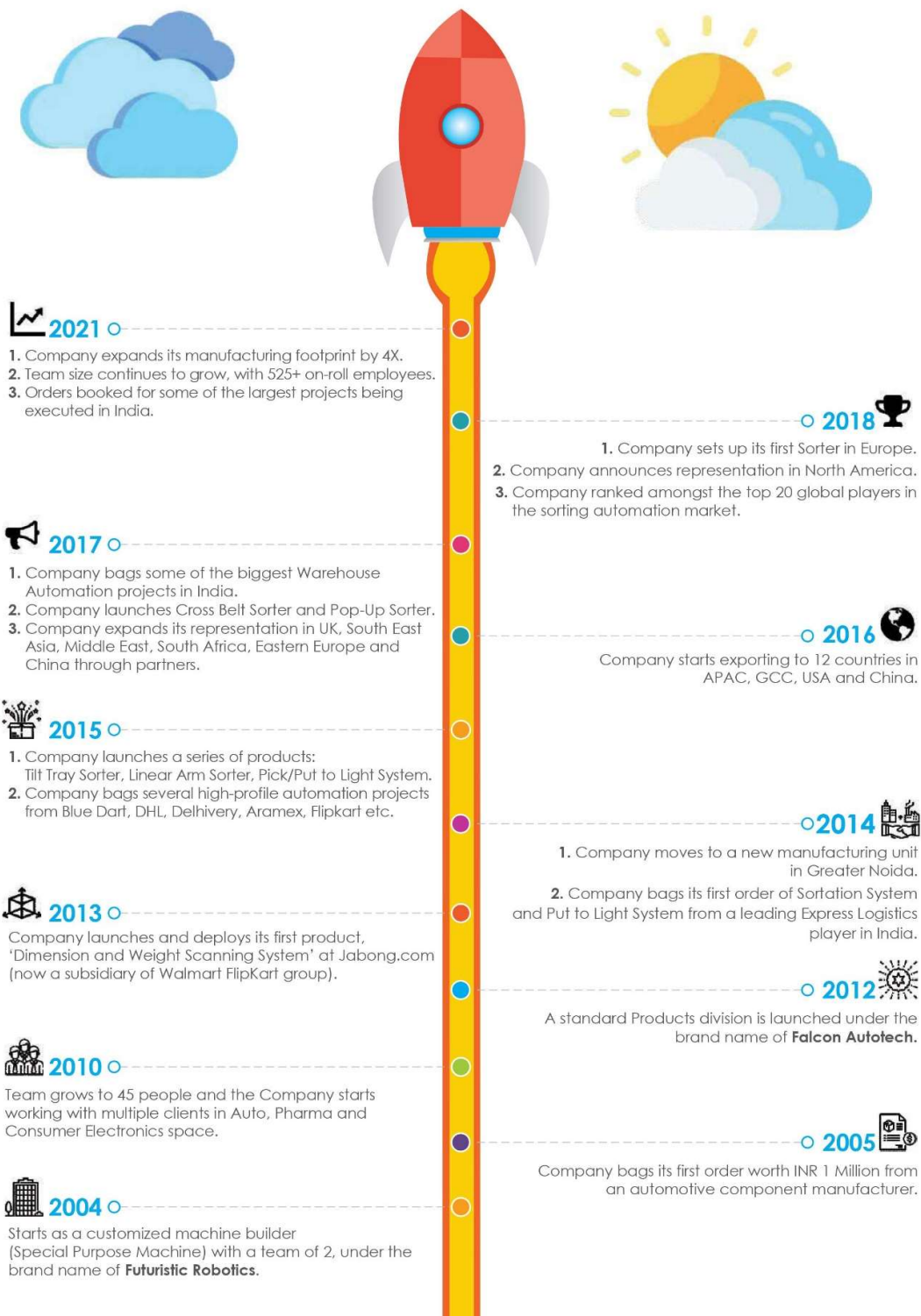
- ❖ Supply Chain Solution Designing and Consultation
- ❖ Hardware Design
- ❖ Electrical Design
- ❖ Electronics Design
- ❖ Software Development
- ❖ Manufacturing

Giving us an unprecedented ability to harness our true potential generated by culmination of Supply Chain Know-How, quality Hardware and flexible Software under one roof.





COMPANY HISTORY





KEY STATISTICS



04

Product Lines



525+

On-Roll Employees



290+

Engineering Team



400000+ Sq Ft

Manufacturing Space



20+

Tier A Clients



15+

Countries with Live Installations



1800+

Total Installed Systems Globally



35

Representations in 4 Continents and 35 Countries



1Million+

Sorts Per Hour of combined Sorting Capacity
installed Globally



OUR PRODUCT PORTFOLIO

01

SORTATION SOLUTIONS



Cross Belt Sorter



Tilt Tray Sorter



Arm Sorter



Swivel Wheel Sorter



Sweep Sorter



Pusher Sorter



Pop Up Sorter

02

PICK/PUT TO LIGHT SYSTEMS



Packet Sortation



Multi-Line Order Consolidation



Order Picking



Store Order Fulfillment



Spare Parts Distribution



Kitting



Multi Order Pick Cart

03

DIMENSION & WEIGHT SCANNING SYSTEMS

STATIC DIMENSION & WEIGHT SCANNING SYSTEMS



Cubizon-R



Cubizon-R THRU



Cubizon-R Cross



Cubizon-R ECO

DYNAMIC CONVEYOR BASED SYSTEMS

OUR 8 MODELS



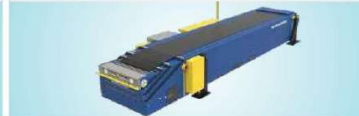


OUR PRODUCT PORTFOLIO

04

CONVEYOR AUTOMATION SOLUTIONS

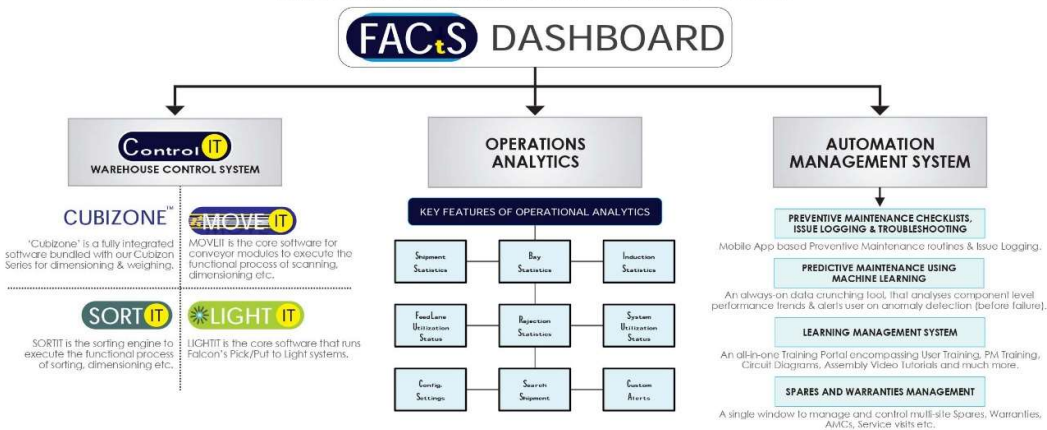
BELT CONVEYORS	ROLLER CONVEYORS	SPECIAL APPLICATION SOLUTIONS
❖ PVC BELT ◆ Straight Conveyor ◆ Spacing Conveyor ◆ Accumulation Conveyor ◆ Strip Belt Merge- 30/45/60 ◆ Solid Belt Merge- 45 ◆ Incline Belt Conveyor ❖ MODULAR BELT ◆ Straight Conveyor ◆ 90 Degree Belt Curve ◆ 180 Degree Belt Curve	◆ Straight Conveyor ◆ Gravity Roller Conveyor ◆ MDR Accumulation ◆ Angled Roller Conveyor ◆ Curved Conveyor ◆ Spur Merge Conveyor ◆ Swivel Wheel Diverter ◆ Pop Up Belt Diverter ◆ Pusher Unit	◆ Aligning Conveyor ◆ One to Two Split Conveyor ◆ Two to One Merge Module ◆ Bulk Distribution-ARB ◆ Bulk Distribution-Telescopic ◆ Bulk Distribution-Electric Arm ◆ Telescopic Belt Conveyor



05

SOFTWARE SUITE

FALCON'S AUTOMATION CONTROL SUITE



— ADD-ON MODULES —



Real Time Video Coding

No-Read Barcodes have always been a challenge in Sorter Applications globally. With Falcon's RTVC feature, we can now easily decode such challenging parcels in real time while sitting remotely.



Deep Learning Image Classification

DLC is an artificial intelligence function that imitates the working of the human brain. It can identify objects such as packets on the carrier and classify them in different categories.



Cloud Sort App/Virtual PTL

It's an android based app used to perform secondary sorting in light assisted put-away operations. It can work as a stand-alone setup or can be integrated with any sorter.



INDUSTRY SOLUTIONS

As a leading player in the intra-logistics automation space, Falcon Autotech continuously strives to improve the operational efficiencies and accuracies for its clients through its domain knowledge and experience in addition to its wide range of products and solutions.

In order to be able to live up to the high expectations set forth by our clients, the team at Falcon Autotech realizes the importance of taking up selective applications in focused Industries and deliver world class projects in return.

FOCUS APPLICATIONS FOR FALCON AUTOTECH

- ❖ Piece Picking and Handling
- ❖ Case Picking and Handling
- ❖ Parcel Handling

FOCUS INDUSTRIES



E-COMMERCE RETAIL



COURIER EXPRESS POSTAL



FASHION DISTRIBUTION



FFP (FOOD, FMCG, PHARMA) DISTRIBUTION



SPARE PARTS DISTRIBUTION

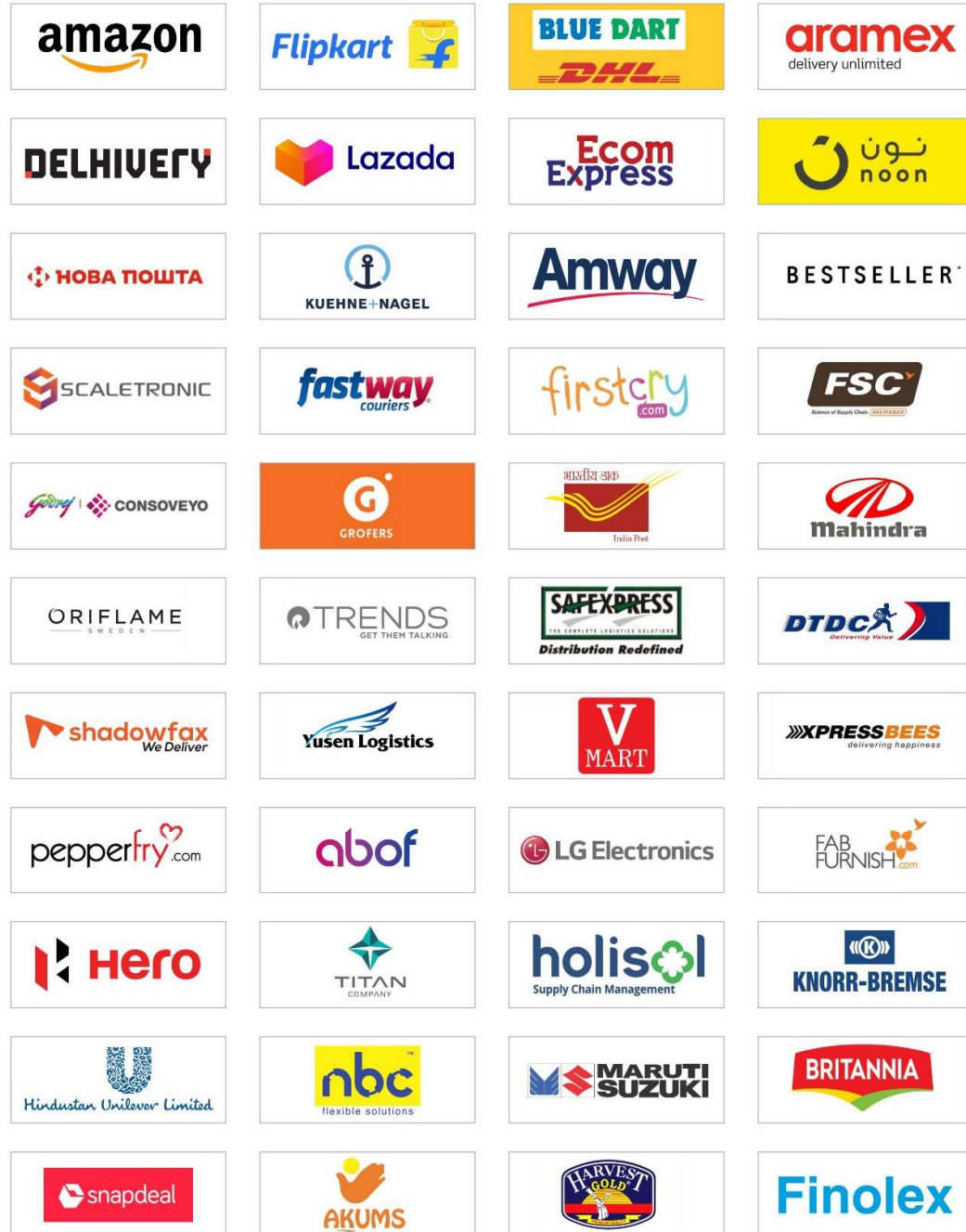


GROCERY DISTRIBUTION



SOME OF OUR GLOBAL CLIENTS

We believe in delivering high quality and standard worldwide. Here is a gist of our esteemed clients across various industry segments.





OUR PRODUCT PORTFOLIO

04

CONVEYOR AUTOMATION SOLUTIONS

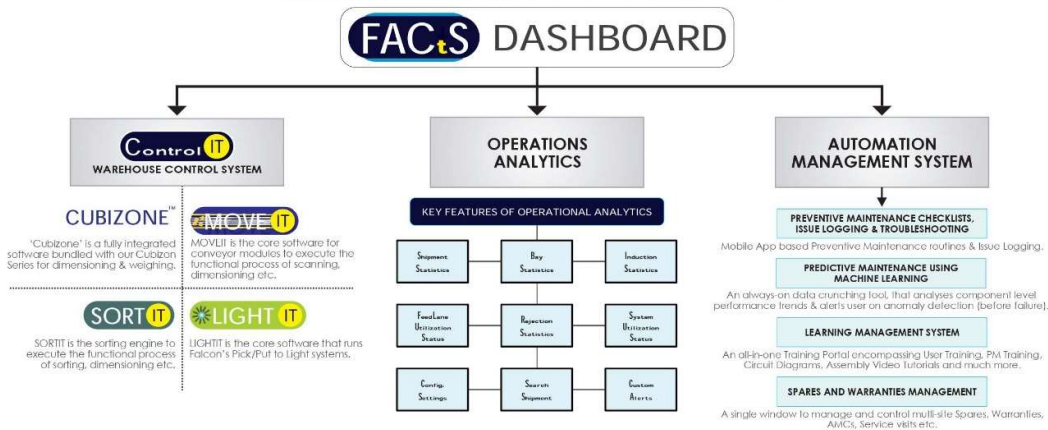
BELT CONVEYORS	ROLLER CONVEYORS	SPECIAL APPLICATION SOLUTIONS
❖ PVC BELT ◆ Straight Conveyor ◆ Spacing Conveyor ◆ Accumulation Conveyor ◆ Strip Belt Merge- 30/45/60 ◆ Solid Belt Merge- 45 ◆ Incline Belt Conveyor ❖ MODULAR BELT ◆ Straight Conveyor ◆ 90 Degree Belt Curve ◆ 180 Degree Belt Curve	◆ Straight Conveyor ◆ Gravity Roller Conveyor ◆ MDR Accumulation ◆ Angled Roller Conveyor ◆ Curved Conveyor ◆ Spur Merge Conveyor ◆ Swivel Wheel Diverter ◆ Pop Up Belt Diverter ◆ Pusher Unit	◆ Aligning Conveyor ◆ One to Two Split Conveyor ◆ Two to One Merge Module ◆ Bulk Distribution-ARB ◆ Bulk Distribution-Telescopic ◆ Bulk Distribution-Electric Arm ◆ Telescopic Belt Conveyor



05

SOFTWARE SUITE

FALCON'S AUTOMATION CONTROL SUITE



ADD-ON MODULES



Real Time Video Coding

No-Read Barcodes have always been a challenge in Sorter Applications globally. With Falcon's RTVC feature, we can now easily decode such challenging parcels in real time while sitting remotely.



Deep Learning Image Classification

DLIC is an artificial intelligence function that imitates the working of the human brain. It can identify objects such as packets on the carrier and classify them in different categories.



Cloud Sort App/Virtual PTL

It's an android based app used to perform secondary sorting in light assisted put-away operations. It can work as a stand-alone setup or can be integrated with any sorter.



INDUSTRY SOLUTIONS

As a leading player in the intra-logistics automation space, Falcon Autotech continuously strives to improve the operational efficiencies and accuracies for its clients through its domain knowledge and experience in addition to its wide range of products and solutions.

In order to be able to live up to the high expectations set forth by our clients, the team at Falcon Autotech realizes the importance of taking up selective applications in focused Industries and deliver world class projects in return.

FOCUS APPLICATIONS FOR FALCON AUTOTECH

- ❖ Piece Picking and Handling
- ❖ Case Picking and Handling
- ❖ Parcel Handling

FOCUS INDUSTRIES



E-COMMERCE RETAIL



COURIER EXPRESS POSTAL



FASHION DISTRIBUTION



FFP (FOOD, FMCG, PHARMA) DISTRIBUTION



SPARE PARTS DISTRIBUTION



GROCERY DISTRIBUTION



INDUSTRY SOLUTIONS

As a leading player in Intra Logistics Automation Space, Falcon Autotech continuously strives to improve the operational efficiency and accuracy for its clients through its domain knowledge consultancy and experience in addition to its wide range of Products and solutions.

In order to be able to live up to the high expectations set forth by our clients, the team at Falcon Autotech realizes the importance of taking up selective applications in Select Industries that lie in the Forte of Falcon Autotech and deliver world class projects in return.

FOCUS INDUSTRIES



E-COMMERCE RETAIL



COURIER EXPRESS POSTAL



FASHION DISTRIBUTION



FFP (FOOD, FMCG, PHARMA) DISTRIBUTION



SPARE PARTS DISTRIBUTION



GROCERY DISTRIBUTION

FOCUS APPLICATIONS FOR FALCON AUTOTECH

- ❖ Piece Picking and Handling
- ❖ Case Picking and Handling
- ❖ Parcel Handling



MEDIA COVERAGE

NO.1 LOGISTICS AUTOMATION COMPANY IN INDIA*



Falcon Autotech has achieved high recognition in the Intra-Logistics Automation world, "Global Logistics Automation Market" report published by Data Bridge Market Research has featured Falcon Autotech among top 20 in global list of Sortation Systems manufacturers. CNBC Awaaz, Awaaz Entrepreneur featured Falcon Autotech as fastest growing Logistics Automation company in India. Besides being featured in many important Daily Newspapers, we have also won the "Best Warehouse Automation Company" of the Year award at Express Logistics Supply Chain Conclave held in Mumbai in the Month of September 2015.

BusinessLine
Home News BUDGET 2017 Markets Companies Money & Banking Economy Info-Social Media Computers & Laptops Mobiles & Tablets Other Gadgets

Robotics powers festival sale impact on productivity for e-commerce brands

JESSU JOHN

October 16, 2016: Between Amazon's 'Great Indian Festival', Snapdeal's 'Utsav Dindi' and Flipkart's 'Big Billion Day' sales, all eyes were on deals and wins earlier this month. However, what's common for most brands executing large impact sales seasons is the need for operational efficiency. Falcon Autotech, an industrial automation solution provider in India, enables efficient fulfilment in the face of a surge of orders experienced by many online retailers during these high demand periods.

Vincent Baid, CEO, Falcon Autotech, explains, "It's the warehouses or fulfilment centres that see a 2-3 times the regular volumes for large festive sale periods."

EFFICIENCY UNPLUGGED

THE FINANCIAL EXPRESS

How automation firm Falcon helping clients deliver products to workers

Industrial automation firm Falcon is helping its clients use robotic carts to pick, place and deliver products from specific locations to workers

Ever wondered how an e-commerce company fulfils thousands of orders everyday—picking up the product from the warehouse nearest the delivery address, packing it properly, labelling it with the right barcode and mandatory receipts, and then ensuring that it reaches the customer in minimal time? While an army of men and machines work round the clock to make it happen,

THE ECONOMIC TIMES SmallBiz
E-commerce | Small Biz | Startups | Finance | Technology | Health | Education | Real Estate | Sports | Entertainment | Lifestyle | Travel | Food & Beverage | Auto | Energy | Environment | Science | Culture | Opinion | Columns

Falcon Autotech looks overseas for growth

By Priyanka Sengupta, ET Bureau | Updated Nov 01, 2015, 08:34 AM IST

PUNE: Falcon Autotech, a Noida-based maker of warehousing automation products, expects international operations to account for about a third of its revenue over the next few years, even as it sees demand for its

THE ASIAN AGE
Delhi | Mumbai | Kolkata | London

Hidden technologies changing our shopping experience

E-commerce sites finally solve the conundrum of sustainable growth and profitability

It is proved that the right product, at the right price, at the right time, is the key to success in e-commerce. However, the success of e-commerce is not just about the product, but also about the technology that powers it. Falcon Autotech, a leading provider of warehouse automation solutions, is helping e-commerce companies solve the conundrum of sustainable growth and profitability by providing them with the right technology.

DECCAN Chronicle

These tech oriented companies are making it smooth for industries to run

TECHNOLOGY, IN OTHER NEWS

Industries still need to cross a lot of hurdles. Technology is playing role in every beat and is innovating to resolve the challenges. Here are few organizations that have made it better for industries in India. Not only have they saved time but are also saving a chunk of money for different industries.

Aaveg manages the entire activity chain including Employee transportation, Leasing to business rental and advising, managing and procuring the overall

PRODUCT CARGOTALK

Cutting edge tech for dimensioning & weighing

The highly accurate and versatile Cubzon-H THRU is equipped with roller bed base, has robust operations along with guaranteed zero-maintenance.

100 YEARS OF GIVING

*Source: Report by Data Bridge Market Research



CLIENT TESTIMONIALS

“ Working with Falcon Autotech has been a very delightful experience for us, BLUE DART being a leader in the express logistics business, we expect the same kind of responsiveness, reliability, commitment & passion from our partners and Falcon has always delivered to their promise on all the solutions such as the Auto Sorter, Weight Dimension Labelling machine, Parcel Locker etc. All the technology interventions have been business critical for us.



Anil Khanna, Managing Director
DHL BLUE DART

“ Falcon is an organisation that has shown great agility and has helped us improve our service capabilities. Their sortation systems have helped us deal effectively with our growing volumes. We have found the Falcon team quite responsive to feedback and quick to incorporate changes into their systems that have helped build better, customized solutions for Delhivery.



Sahil Barua, CEO
DELHIVERY

“ Falcon Autotech has been efficient and proactive, particularly in meeting deadlines and their technical advancements. Falcon Autotech strike just the right balance between being a flexible, easy to deal with firm yet having all the key strengths and attributes. Every member of the team is honest, reliable, responsive, personable and very knowledgeable. I wish them good luck for all the future endeavours.



Sanjeev Saxena, Director
ECOM EXPRESS

“ I believe you guys earned it with your hard work and dedication. I believe the products you have designed demonstrate innovation and lateral thinking. Also the management has a flexible and the customer centric approach. It has been a delight to work with you and the team.



Amitava Saha
CEO, firstcry.com



3. Falcon's Experience and Achievements in Sortation Space Globally

- Ranked among **Top 20 Sortation System Suppliers** globally. Only company from APAC/EMEA Region.
- Currently possess one of the **World's largest portfolio in Sortation Technologies**: 7 In-house technologies.
- Total installed capacity of **6 Million Parcels per day** worldwide.
- Only company to be able to offer a **Fully Integrated AMS**.

Born2Invest
Start A Report Or Project

Incas-SSI Schaefer merger creates leading player in Italy's intralogistics market

TOP STORIES • **CROWDFUNDING** • BUSINESS • FINANCE • CRYPTO • COMMODITIES • LUXURY • TOUCHPOINTS • SO

Although, the transportation management segment of the logistics automation industry is expected to have a higher CAGR. This is due to factors associated with the growing need to deliver goods in a timely manner and to reduce transportation costs.

Moreover, automotive will be the leading segment of the industry from 2018 to 2023, with an increasing demand for automated production systems and flexible logistics systems used in manufacturing and supply of vehicle fleets.

The MarkestandMarkets research report has identified SSI Schaefer as one of the key players in the logistics automation market worldwide during the forecast period, along with Dematic Corporation (US), Murata Machinery, Ltd. (Japan), Jungheinrich AG (Germany), and Falcon Autotech (India), among others.

This year's largest market for logistics automation is estimated to be North America, according to the research report. Regardless, Incas and SSI Schaefer's merger could help them compete in their main market.

(Featured image by DepositPhotos)

0 SHARES

Facebook Twitter LinkedIn WhatsApp Email

RELATED TOPICS: #AUTOMATION #BENNO REICHMUTH #ERMANNO RONDI #FEATURED #INCAS SPA #INTRALOGISTICS MARKET #MARKETSANDMARKETS #SSI SCHAEFER GROUP

Not secure | worldindustryinsights.com/30269/global-parcel-sortation-system-market-analysis-size-share-growth-trends-and-forecast-2019-2026-siemens-vanderlande-beumer-group

Global Parcel Sortation System Market Analysis, Size, Share, Growth, Trends, and Forecast 2019-2026: Siemens, Vanderlande, Beumer Group, Intelligrated, Bastian Solutions, Fives

BY ANN.CASTRO ON AUGUST 12, 2019

The global "[Parcel Sortation System market](#)" research report is a pervasive research report that touches the most vital parts of the Parcel Sortation System market that is necessary to be grasped by a professional or even a layman. The market research report enlightens one regarding few of the important aspects such as an overview of the Parcel Sortation System product, the growth factors enhancing or hampering its development, application in the various fields, major dominating companies, genuine facts, economic situation, and geographical analysis. The research report endows the data concerning the dynamics that propel the growth as well as the supply and demand chain of the product on a global basis.

Click here to access the report: www.reportsbuzz.com/request-for-sample.html?repid=50822

The international and global stand of the Parcel Sortation System market is also briefly mentioned in the research report based on the performed statistical and thorough market analysis. The information mentioned in the research report gives a qualitative and quantitative view of the overall market. The statistical analysis of the market helps analyze the supply, demand, production, maintenance, and storage costs of the product. The data regarding some of the prevalent players Siemens, Vanderlande, Beumer Group, Intelligrated, Bastian Solutions, Fives, Dematic, Interroll, Muratec, Invata Intralogistics, Viastore, Okura Yusoki, GBI Intralogistics, Equinox Mhe, Pitney Bowes, Falcon Autotech, OCM SRL, Solystic, Bowe Systec, Intralox are also detailed in the current case study.



4. Reference projects

Falcon has a strong legacy in **Warehousing Automation** solutions and references-

1. Expertise in Piece Picking and Handling, Case Picking and Handling & Parcel Handling
2. Lifecycle services (maintenance, spares supply chain, support).
3. Full **in-house** expertise (Hardware/Software).
4. Turn-key **tailored** solutions.

The references list presented below focusing on Sortation Solution –

4.1 Project 1- (Client – Courier Express Parcel, Dubai & Jeddah)

This Solution is designed to handle the bulk volumes through Launchpads and induct them into Falcon's Linear Arm Sorter. This system is designed for a throughput of 3000 parcels per hour equipped with automatic barcode scanning, dimensioning, weighing and image capture capabilities to into a total of 480 end destinations with a combination of primary and secondary sortation system. The secondary sortation is achieved by integrating put to light system.

Solution Specifications –

- Throughput: 3000 PPH
- End Destinations: 480 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Linear arm sorter.
- Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- WCS Software System.

Site Picture –





4.2 Project 2- (Client – E-Commerce, India)

Use Case – Destination Sorting of Packed Parcels.

In 2019, Client was looking for a potential automation partner for design and development of a new automated sortation system for B2C parcels. The system should be able to provide maximum uptime with reduced dependency on skilled manpower, and space optimization.

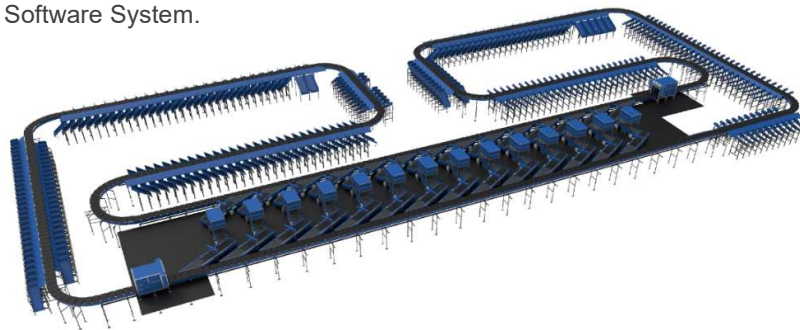
The customer chose Falcon Autotech based on its unique design which could cater to all their pain points, capability of seamless integration with WMS and life cycle support services.

Solution Specifications –

- Throughput: 27,600 PPH
- End Destinations: 410 Direct Outputs
- Building Size: 200,000 Sq Ft

Key Technology Modules –

- Bulk Infeed Conveyors.
- ARB based Volume Distribution System
- Integrated Presort System.
- Irregular Ejection System.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- Smart Sliding Chutes for Direct bagging and Cage Sorting.
- Bag take out system
- WCS Software System.





4.3 Project 3- (CEP Client, Riyadh)

In 2019, customer selected Falcon Autotech as a preferred supplier for its airport hub to supply linear cross belt sorter for processing of parcels arriving via Air from different states and countries to distribute them locally. The customer chose Falcon Autotech based on its strong track record of success in providing intralogistics automation solutions, optimized design, software integration capabilities and life cycle support services.

Solution Specifications –

- Throughput: 4800 PPH
- End Destinations: 52 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- ARB Conveyor.
- Automatic Label Applicators.
- Linear Cross Belt Sorter.
- Specialized Chutes for Gentle Parcel handling.
- FOCR Engine.
- WCS Software System.

Site Picture –





4.4 Project 4- (CEP Client, UK)

This Solution is designed to handle a volume of 7200 parcels per hour. The system is equipped with three infeed conveyors integrated with automatic label applicator before parcels enter the sortation system. The parcels are sorted using Falcon's Loop Cross Belt Sorter equipped with automatic barcode scanning, dimensioning, weighing and image capture capabilities. The sorter is installed on the mezzanine floor and sorts directly to 58 end destinations.

Solution Specifications –

- Throughput: 7200 PPH
- End Destinations: 58 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- WCS Software System.

Site Picture –





4.5 Project 5- (CEP Client, Sydney)

Solution is designed for handling a throughput of 16,000 parcels per hour with the help of Falcon's Loop Cross Belt Sorter. The system consists of 2 feeding zones with a total of 10 feedlines. Sorter design enables the van drivers to directly drop the parcels at the dock doors. It has a total of 369 end destinations that are achieved with a combination of direct drops and PTLs. System is integrated with 5 side automatic barcode scanning, weight and volume measurement and automatic detection of oversize and overweight parcels.

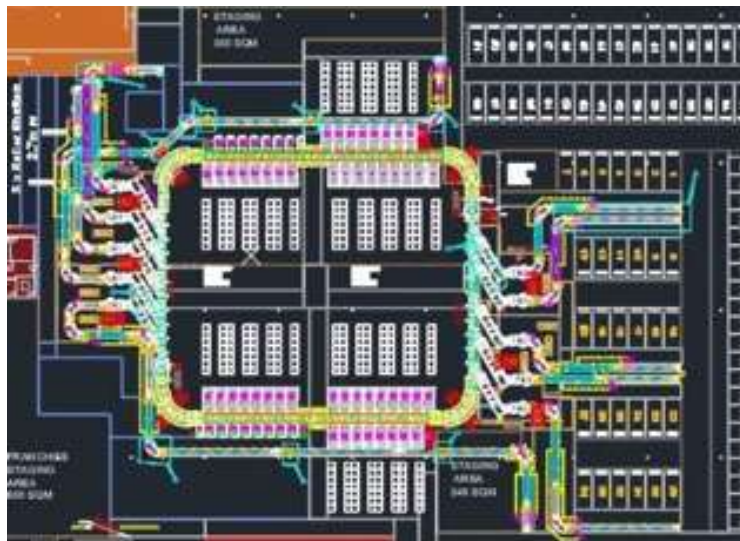
Solution Specifications –

- Throughput: 16000 PPH
- End Destinations: 369 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- 2 Induct zone.
- 5 side Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- Automatic detection of oversize parcel.
- Loop Cross Belt Sorter.
- WCS Software System.

Site Picture –





4.6 Project 6- (Ecommerce Client, India)

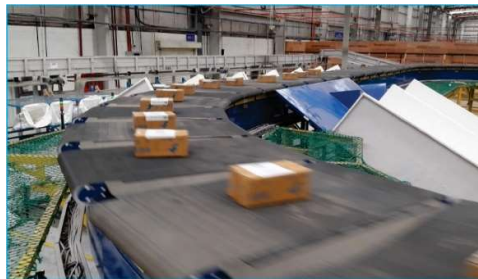
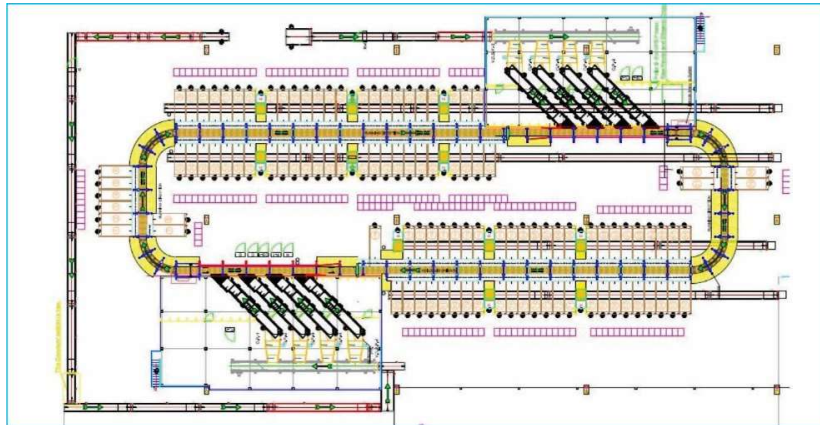
Use Case - Destination Sorting of Packed Shipments

Solution Specifications –

- Throughput: 24,000 PPH
- End Destinations: 110
- Building Size: 200,000 Sq. Ft

Key Technology Modules - Loop Cross Belt Sorter with a cell size of 900 x 500 mm

Layout and Site Pictures -



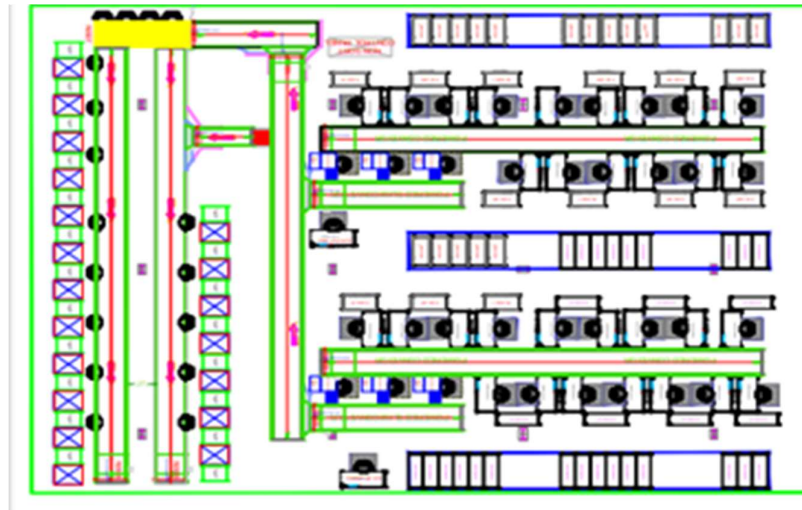
Layout & Pictures_ Project 1



4.7 Project 7- (E-commerce, India)

Solution is designed to handle the parcels Inbound and Outbound for packaging and sorting. Types of conveyors have used in this solution are as follows :

- Belt Conveyor
- Roller Conveyor
- Goose Neck Inclined





5. Handled Parcel Spectrum

5.1 Product size Conveyable on the entire system.

Max Length	mm	1200
Max Width	mm	600
Max Height	mm	600
Max Weight	Kg	35
Min Length	mm	150
Min Width	mm	150
Min Height	mm	10
Max Weight	Kg	2

5.2 Parcels/Totes to be loaded on Conveyor/Sorter shall have the following characteristics:

1. Centre of Gravity of items must not move during conveyance or sorting.
2. Item must not have magnetic content; otherwise behaviour of parcel cannot be guaranteed.
3. Liquid or fragile material, to avoid breaking, spillage or leakage, such as wine bottles, metal cans of paint are designated as non- conveyable items.
4. Parcels shall be perfectly and safely packaged: protrusion or open surfaces are not allowed.
5. Plastic ropes shall be perfectly adherent to the surface of the package.
6. All items with the risk of being damaged during the transport on an automatic conveying/sorting system or damaging the system. They must be robust enough to avoid disintegration of container material and loose of contents in the conveying/sorting process.
7. Item packaging shall have enough grip to be handled on the belts during the acceleration and referencing phases.
8. Items shall not have slippery surfaces and must be able to withstand acceleration of the items on the belt during the start-stop phases (accelerations up to 0.5 g shall be assured without any sliding or tumbling of the items on the belt conveyor).
9. The parcels must have at least one flat and regular surface providing enough stability during conveyance.
10. All shapes are permitted except spherical, cylindrical, or alike unstable items & shapes.
11. All usual packaging materials are permitted (including paper, carton, plastics, plastic foil, rope, tape, textile, and wood)



5.3 Parcel not loadable on the system

All products that are not within the range as described here, are considered non- conveyable products, and must be taken out of the main sorter flow by the operators.

1. Unstable items with a risk to roll or tumble on the sorting system, such as spherical or cylindrical items
2. Items that have a spherical or cylindrical shape.
3. Items that are packed in material that can damage the conveyors or the sorter.
4. Items that have sharp points (e.g. Nails) or sharp edges, that can damage the conveyors or the sorter.
5. Fragile parcels with contents not sufficiently secured
6. Items that have been classified as dangerous are designated.
7. Wet items are designated.
8. Items with anti-slip treatment.
9. Items with protruding parts.
10. Items with sharp edges.
11. Inadequately packed items that could be damaged during automatic transportation.
12. Electrostatically loaded items.
13. Loose parts on loads and load carriers, such as adhesive tape, stickers, slips of paper, straps, wrap foil etc. are designated as non- conveyable items.

5.4 Types of Products

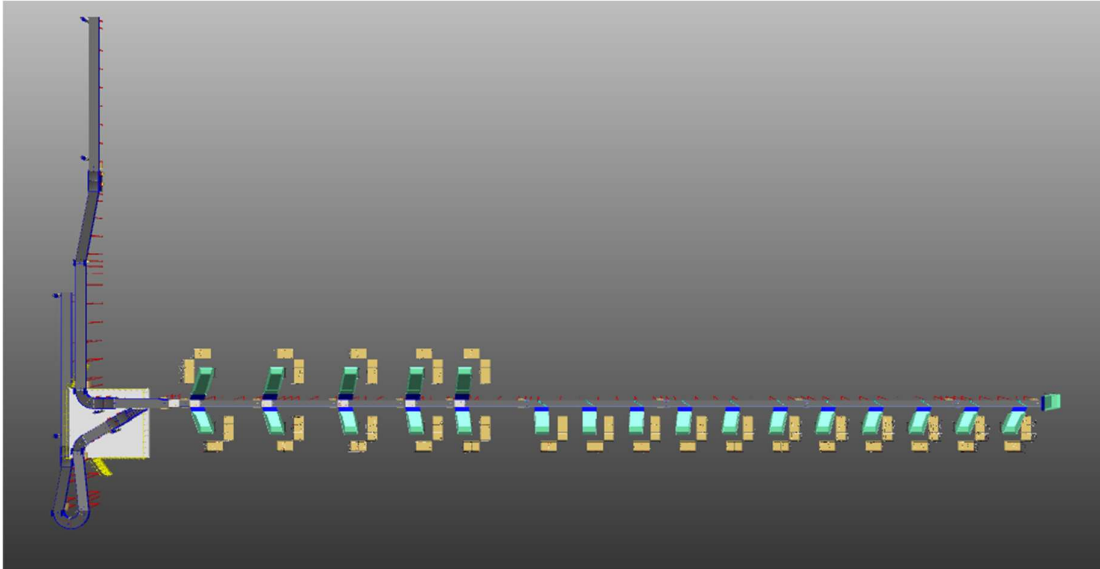
Bags/Semi Large Shipment	Max: L 1200 mm X W 600 mm X H 600 mm Min: L 150 mm X W 150 mm X H 10 mm



6. Proposed System Description

6.1 Objective

The purpose of this proposal is to present the design, manufacturing, installation, commissioning, testing and acceptance testing of the complete combination of SWEDI & ARM sorter system to handle Parcel as per specifications shared by Flipkart.



6.2 Summary of the System

1. **Infeed:** Manual loading of Bags & Semi-large shipment is done by Operators on both loading conveyor.
2. **Parcel Scanning:** Bags & Semi-large shipment are manually scanned with the help of hand-held scanner & then inducted through Semi-Automatic Induct lines.
3. **Outbound Sorter:** Bags & Semi-large will travel to outbound sorter after getting aligned by SWEDI unit & then will be diverted with the help of SWEDI & ARM diverter into collection type chute.
4. **Sort Fail:** At the end of the line sort fail chute is provided.

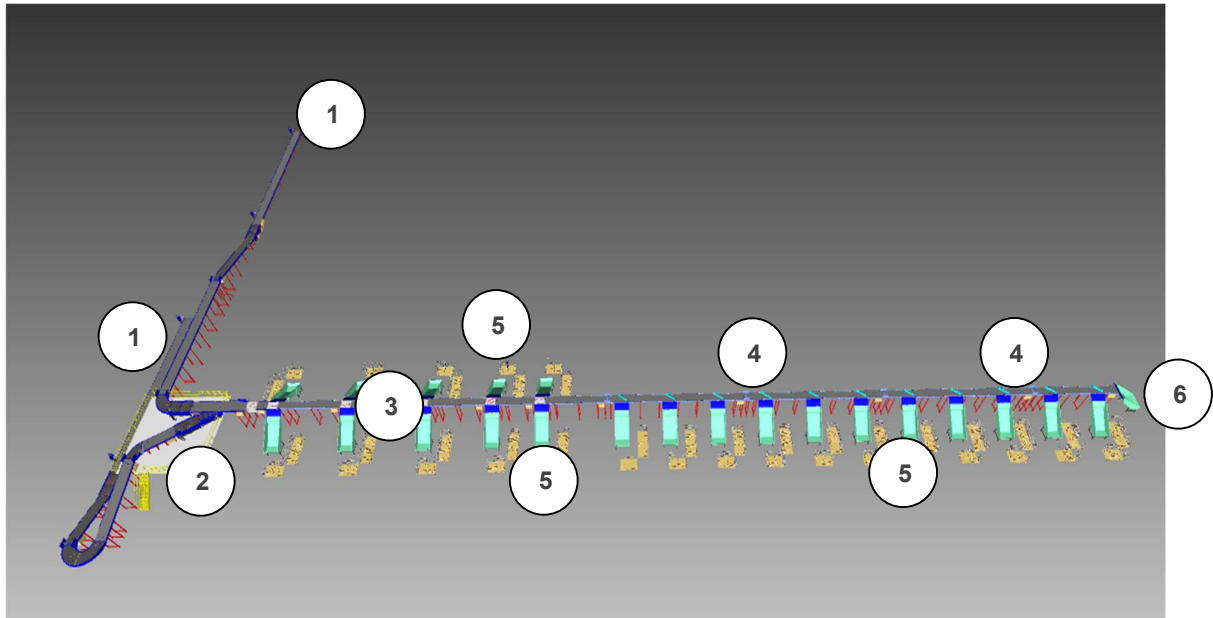
6.3 Main benefits of the proposed solution

The proposed solution has the following advantages:

1. High operational throughput
2. Maximized sorting capacity throughout at dispatch sorter to meet the requirement.
3. Low occupancy of floor space in the building
4. FALCON's system can adapt with changing business requirements by adjusting its speed to match the operational throughput requirement, thereby leading to Power Savings and reduced system Wear & Tear



7. Layout View



Legend

- 1 – Loading Conveyor
- 2 – Induct lines
- 3 – SWEDI units (6 Nos)
- 4 – ARM Diverter (11 Nos)
- 5 – Collection Type chute
- 6 – Sort fail chute



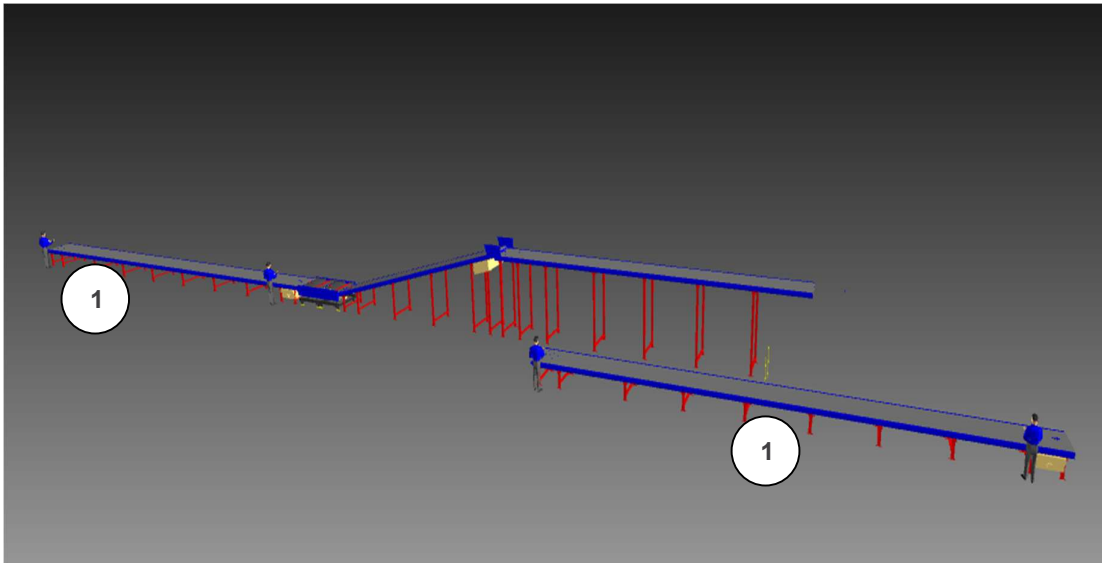
8. Proposed System Capacity

8.1 System Capacity

Activity	Parcel per hour
SWEDI Diverter	1500
ARM Diverter	1500

9. System description

9.1 Loading conveyor

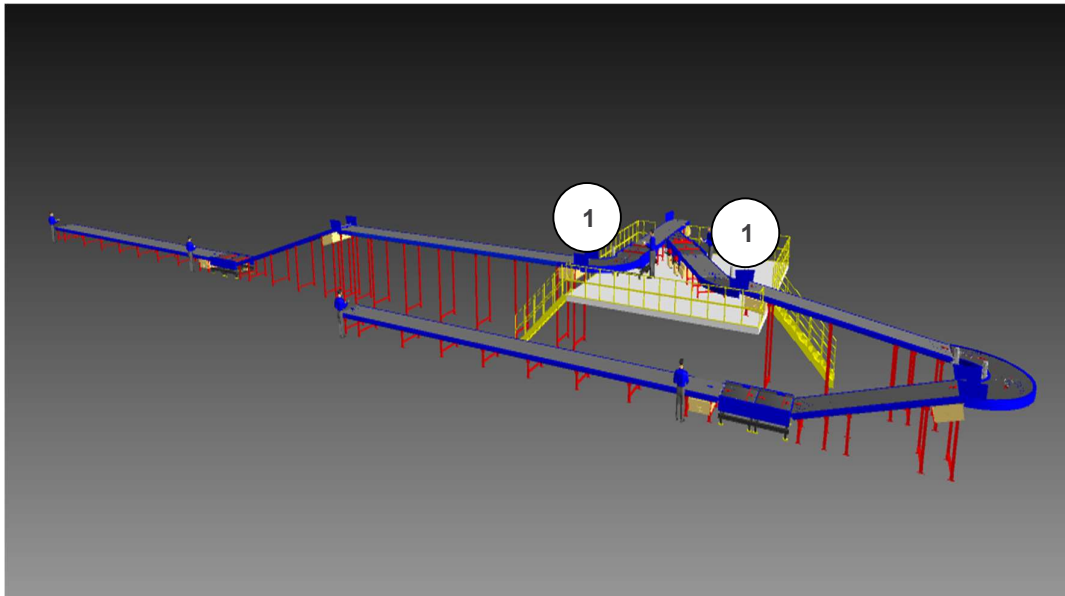


Legend

1 – Loading conveyors



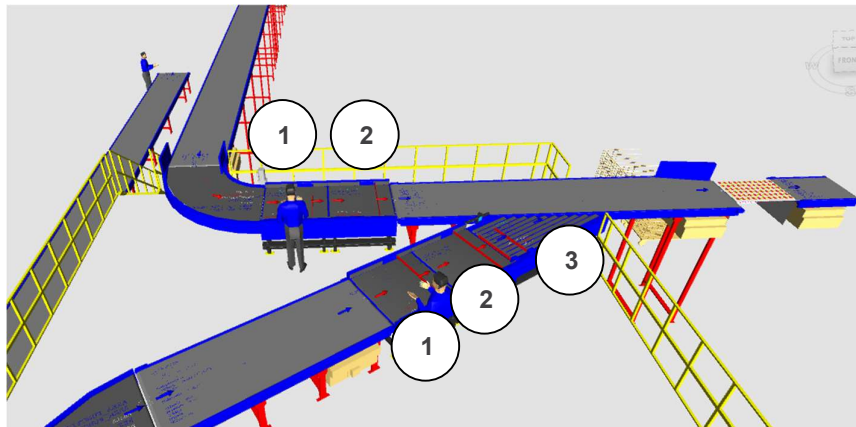
9.2 Manual Scanning



Legend

1 – Handheld manual scanning

9.3 Parcel Induction



Legend

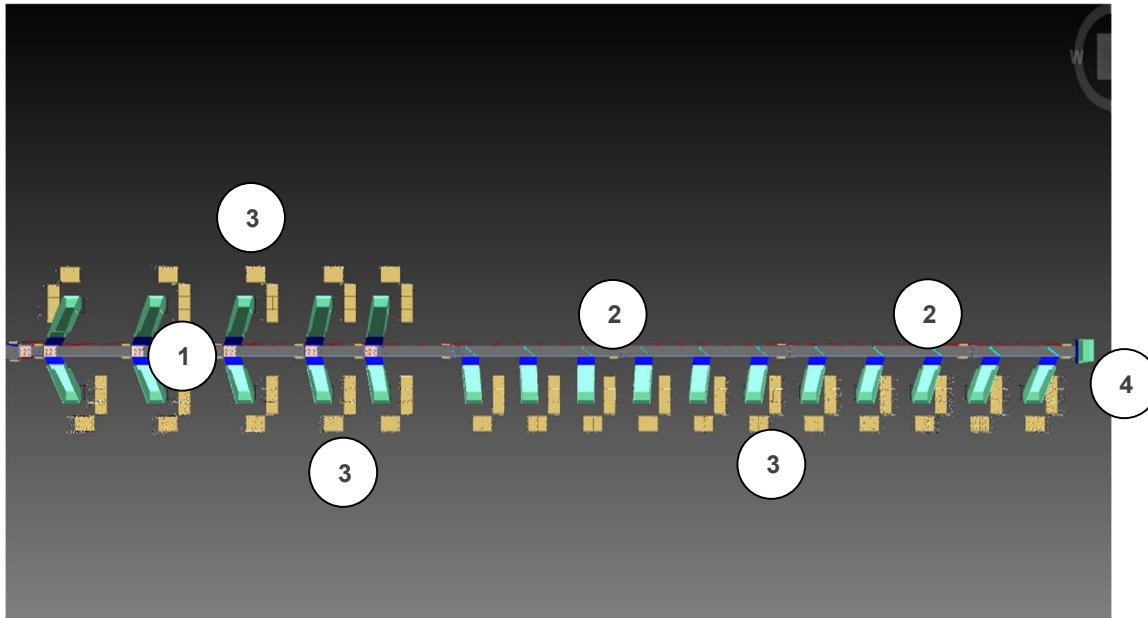
1 – Scanning conveyor

2 – Buffer conveyors

2 – Intelligent merge



9.4 Outbound Sorting



Legend

- 1 – SWEDI diverter
- 2 – ARM diverter
- 3 – Collection chute
- 4 – Sort fail chute



10. System Components

10.1 PVC Belt Conveyor

Falcon's Belt Conveyors are modular & robust in design. MS profile is used to build conveyor frame.

Some Salient Features of Falcon's Belt conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
MOC of Belt	Type	PVC
Belt Thickness	mm	3
Belt Finish	Type	Matt Finish
Belt width	mm	1200
Belt Joint Type	Type	Vulcanized, Endless Belts
Belt Make/Model No.	Make	FORBO/DERCO
Chassis	mm	MS, 3mm Thick, Powder Coated-75 Microns
Idler Roller Pitch on running side (Top)	mm	480
Idler Roller Pitch on return side (Bottom)	mm	1200
Idler Roller Specs	Spec	Dia.50.8, shaft-12 mm, bearing-6301
End Rollers	Spec	Dia 81.2 x 2.85 Thick, 30mm Shaft Dia, & Bearing Size-UCFH206D1, MOC-EN9, Zinc Plated -25 microns
Drive Rollers	Spec	OUTER Dia 178 PIPE Dia 168x5 Thick, 40mm Shaft Dia, & Bearing Size-UCF208D1, MOC(SHAFT)-EN9, NEOPRENE RUBBER COATING



Tensioning Rollers	Spec	Dia 81.2 x 2.85 Thick, 25mm Shaft Dia, & Bearing Size-UCT205D1, MOC-MS, Zinc Plated -25 microns
Type of Drive	Type	Rubberized
Angle of wrap/Contact - Belt with Drive pulley	Deg	>180 deg
Motor Shaft	MOC	EN9 Shaft material,
Conveyor Under guarding	Spec	MOC-MS, 0.8 to 1mm thickness, Steel Stiffeners at every 600mm to avoid the sagging due to self-weight
Load capacity per unit length of Conveyor	kg/m	50
Drive Power Rating	kW	Motor Rating depends on Length of the Conveyor
Type of Motor	Type	AC Geared Motor
Type of Drive System	Type	Direct Shaft Mounted Motor/ Flange Mounted
Type of Drive	Type	Chain drive/Torque Arm type
Gear Motor	make	Sew/Nord
Drive	make	Lenze/Siemens/Omron/Allen Bradly
Insulation Class	Type	Yes
Conveyor Speed	m/s	Variable speed to meet TPH requirements
Type of Mounts	Type	Fixed Heights Legs with Grouting Provisions

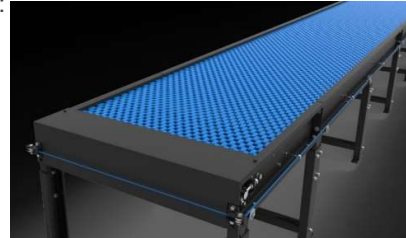


10.2 Plastic Modular Belt Conveyor

Falcon's Belt Conveyors are modular & robust in design. MS profile is used to build conveyor frame.

Some Salient Features of Falcon's Modular Belt conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Plastic Modular Belt Material	Type	POM with PBT Pins, Pitch-25.4 mm, Pin Dia- 5mm
Plastic modular Belt Make/Model No.	Make	Movex/XP254 CR
Sprocket Size & No. of Sprockets on Drive End & Tail End	Type	5/146.4 mm
Sprocket material	MOC	POM
Chassis	mm	MS, 3mm Thick, Powder Coated-75 Microns
Wear Strip Material	MOC	UHMW PE1000
Wear Strip Pitch	mm	150 (Strip Width - 40mm)
Return Roller type	Type	DIA- 50 mm/20303
Return Roller Pitch	mm	800
Driven Shaft	Type	MOC-EN9 Solid Shaft hardened to BHN 240, Size-60SQ MM, Zinc Plated -25 microns, Bearings-UCF-210D1
Duplex Sprocket	MOC	MS
Duplex Chain	MOC	MS
Drive Shaft	MOC	EN 24
Conveyor Under guarding	Type	GI SHEET (1MM THICK)
Type of Guides	Type	Steel guides at 50mm/400mm height basis the position of conveyor on both sides from Conveyor Belt Surface.



10.3 Swivel Wheel Sorting Unit - Falcon

Falcon's Swivel wheel divert is used in system for the smooth transfer of products.

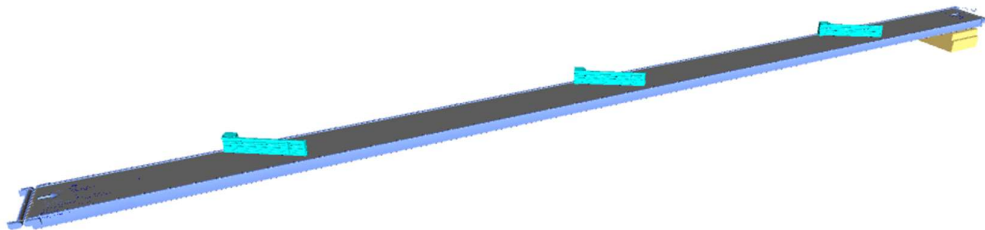
Some Salient Features of Swivel Wheel Divert:

- Low Noise
- Maximum uptime.
- Electric based rotation of wheels.
- Minimal maintenance
- High safety standards.



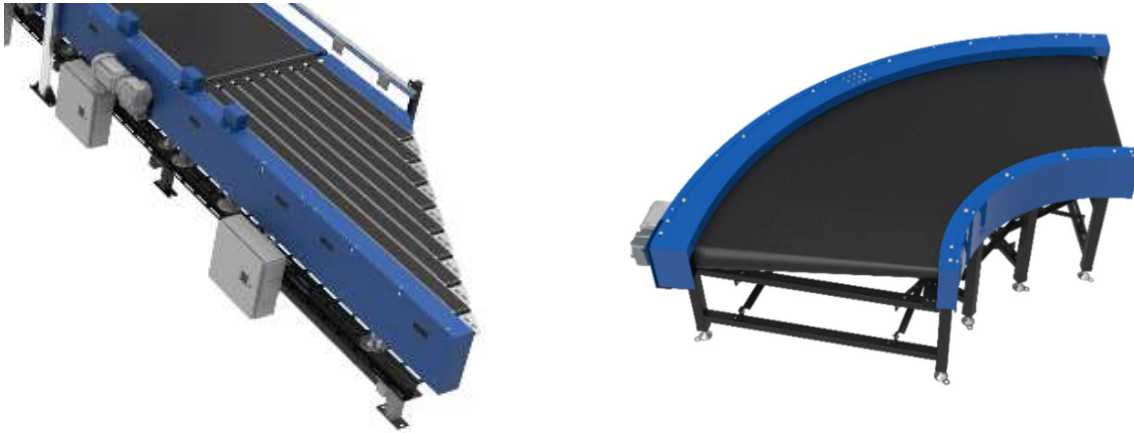
Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Type - Equipment	Type	Merge/Sorting
Type - Operation	Type	Sorting
Mechanism for Swivel	Type	Electric servo motor
Roller drive	kW	Powered smart wheel
Roller pitch (L x W)	Type	110 x 120
No. of divert sections	Yes/No	1 or 2
Type of Mounts	Type	Fixed Heights Legs with Grouting Provisions

10.4 ARM Sorting Unit





10.5 Powered Belt Merges & ZTR Curves (Pic for ref.)



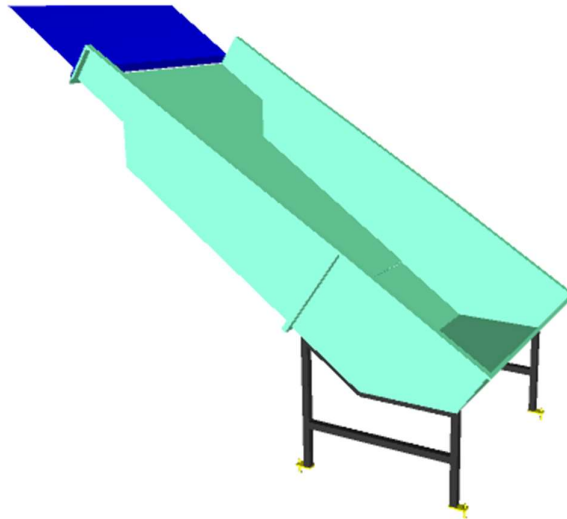
10.6 Handheld Barcode Scanners

Both induction lines area will be equipped with a Handheld wireless 2D Barcode scanner This scanner works as a backup for Auto scanner as well to ensure higher system redundancy.





10.7 Collection Type Chute

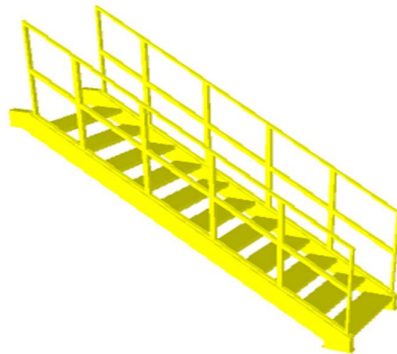


10.8 Operator's Platform





10.9 Stair case



11. Proposed System Technical Details/BOM

11.1 Mechanical Equipment

S.no.	Qty.	Description	Value
1	1	Conveyors 13 X Powered belt conveyor ~ 118 m 01 X Modular conveyor~ 6.7 m 04 X Modular conveyor~ 64 m (Arm sorter) 02 X Scanning conveyor 03 X 90 deg Curves 01 X 60 deg Curves	Width ~ 1210 mm
2	1	Diverter 06 X SWEDI units 11 X ARM units	Included
3	1	Destination/Chute 21 X Collection type chute 01 X Sort fail chute Tower lamp (As per req.)	Included
4	1	Steelwork 01 X Operator's Platform ~ 86.9 sqm 02 X Staircase	Included
4	1	Scanning Unit 02 X Handheld barcode scanner	Included
5	1	Standard Flipkart Safety Equipment	Included



11.2 Electrical Equipment

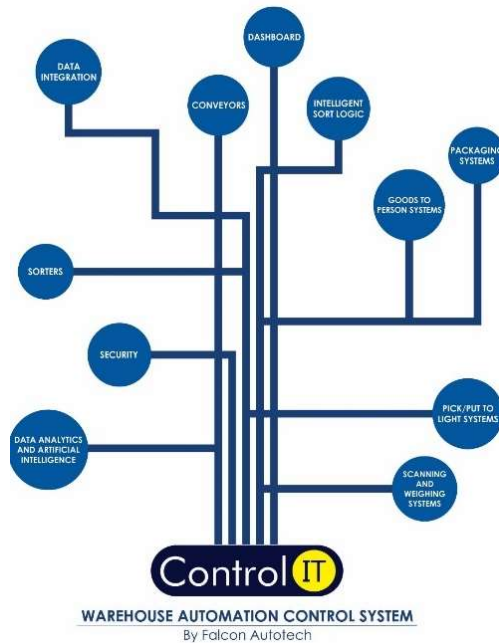
Electricals			
1	1	Consists of Main Control Panel Network Switches Field Cabling	Included

11.3 Control System

Components			
2	1	Controls, IT and Wiring Software Package Control Panels Sorter Control Electrical & Control Hardware Switches Drive panels Operator Panels Internal & Field Wiring	Included



12. Falcon's WCS CONTROLIT



Key Values

Supports Multiple Use Cases

- Order Consolidation, SKU wise Sorting, Route Level Sorting, Transporter Wise Sorting and many more

Supports dynamic and static Sort Logic Definition and unlimited Sort Logic Wave definitions

Supports various peripheral Equipment Inputs and Outputs

- Automated Scanners, Manifest Printers, Label Applicators and many more

Ability to define Custom Business Logics

- Cut Off Timings, Quantity/Weight and Volume Thresholds etc

Security

- User profiling with Role Right Mapping
- Database Encryption
- Secured and Protected Communication between WMS and WCS layers

Supported Protocols

API	FTP	FILE UPLOAD / DASHBOARD	ODBC
WEBSERVICES	AMAZON S3 SERVER	FIREBASE	Any Customer Protocol



13. High Availability Server Specifications (In Flipkart's Scope)

S. No	Configuration	Role	Qty	Purpose
1	Dell PowerEdge T440 (Chassis with up to 8, 3.5" Hot Plug Hard Drives) Tower Server Trusted Platform Module 1.2 Intel Xeon Silver 4110 2.1Ghz, 8C/16T, 9.6GT/s 2UPI, 11M Cache, Turbo, HT (85W) DDR4-2400 40 GB (2 x 16GB, 8 GB) RDIMM 2666MT/s Dual Rank 2 x 1TB SATA 7.2K 3.5" 6Gbps Fully assembled (Hot Plug) Hard Drive PERC H330 Integrated RAID Controller DVD +/-RW, SATA, Internal Single Hot-Plug Redundant Power Supplies (1+0) 495W Dual-Port 1GbE On-Board LOM iDRAC9 Express 3Yrs Basic Warranty - NBD	Node 1 - Prod Server	1	Failover Cluster is configured on this system. If this node fails, then the VM shifts from this node to another and vice versa
2	Dell PowerEdge T440 (Chassis with up to 8, 3.5" Hot Plug Hard Drives) Tower Server Trusted Platform Module 1.2 Intel Xeon Silver 4110 2.1Ghz, 8C/16T, 9.6GT/s 2UPI, 11M Cache, Turbo, HT (85W) DDR4-2400 40 GB (2 x 16GB, 8 GB) RDIMM 2666MT/s Dual Rank 2 x 1TB SATA 7.2K 3.5" 6Gbps Fully assembled (Hot Plug) Hard Drive PERC H330 Integrated RAID Controller DVD +/-RW, SATA, Internal Single Hot-Plug Redundant Power Supplies (1+0) 495W Dual-Port 1GbE On-Board LOM iDRAC9 Express 3Yrs Basic Warranty - NBD	Node 2 - Prod Server	1	Failover Cluster is configured on this system. If this node fails, then the VM shifts from this node to another and vice versa
3	Intel® Xeon® processor E3-1200 v5 product family, 16 GB RAM, 1 TB * 2HDD, RAID 1, iDRAC8 Enterprise (upgrade), 3 years Onsite warranty.	Domain Server	1	Hyper V will work with Domain System So that we required Domain Server Which will manage the nodes
4	Synology storage with 4 TB HDD	Network Storage Box 1	1	For Creation of Clustered Share Volume, VM will be created on this BOX
5	Synology storage with 4 TB HDD	Network Storage Box 2	1	Backup & Replication of CSV
6	Power Backup	Power Backup	3	UPS for the Servers & Network Boxes
7	Monitor - 19.5 inch	Monitor	1	Monitor
8	KVM Switch	Switch	1	To switch over the display between servers
9	Windows Server 2016 Standard R2 or Higher	Windows License	3	OS



14. Key Component Makes

Component	Make
PLC & RIO	Siemens
VFD	Siemens/Omron/Lenze/AB
Ethernet switch	Siemens/Phoenix Contact
SMPS	Omron/Selec/Meanwell
Slim relays	Phoenix contact
Terminals	Wago
Tower lamp + Hooter	Leuze
Switchgear	Schneider
Geared Motors	SEW
Volume Scanner	N/A
Cameras	N/A
Weighing Scale	N/A

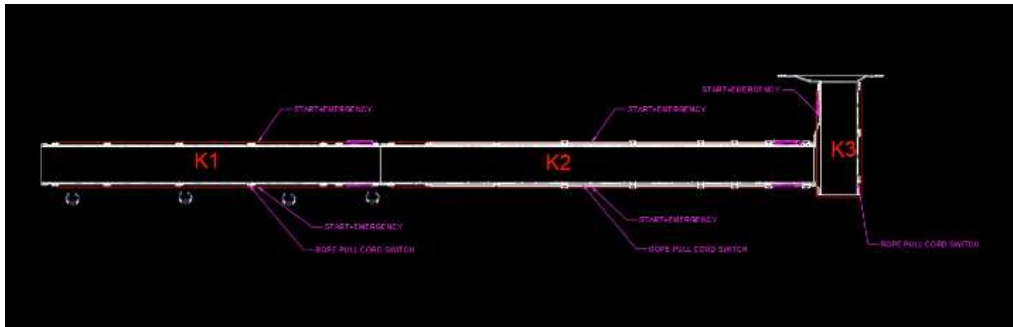


15. Safety

15.1 Push Buttons

Push Button

Emergency Push Buttons will be provided at all the points of human interaction on the system as per Falcon Standard design parameters.



15.2 Program Management

For this program, proposed approach covers the following aspects:

1. Creation and monitoring of the project plan.
2. Communication with Flipkart
3. Scheduling of the resource management.
4. Management of the risks and opportunities.
5. Management of the list of anomalies or reservations.



16. Commercial Terms

16.1 Price Sheet

S. No	Component	Set	Price (INR)
1	SWEDI Diverters	1	₹ 95,62,740
2	ARM Diverters	1	₹ 91,09,000
3	Conveyors	1	₹ 1,64,29,036
4	Collection Type Chutes	1	₹ 42,76,859
5	Operators Platform	1	₹ 13,29,436
6	Software Packages	1	₹ 6,00,000
7	Toolkit	1	₹ 79,800
8	Installation & Commissioning	1	₹ 20,39,344
9	Packaging & forwarding	1	₹ 8,15,737
10	Project Management	1	₹ 11,06,049
Total (INR)			₹ 4,53,48,001

The Total Price is to be understood.

- Freight- Excluded
- GST will be charged extra @ 18%
- Including Packing and Forwarding
- Installation and Commissioning: Included
- Price is valid for 60 days from the date of proposal.

Our Price does not include taxes, and duties, import duties, charges, fees, taxes etc. charged by authorities outside India for deliveries and services. Same shall be borne & paid by client itself.

*Material Unloading & Laydown area is in customer scope

16.2 Payment Terms

The Supplier Quotation is based on the following payment schedule:

Stage 1: 40% Advance along with PO

Stage 2: 50% + Taxes Against dispatch of the material

Stage 3: 10% against Installation



16.3 Delivery timeline

16 – 18 weeks for dispatch of the material from date of PO, Advance & DAP

5 – 6 weeks for Installation

17. Warranty Period

Falcons offered System comes with a standard warranty of 1 year (Starts from the date of beneficiary use) and post completion of warranty, customer can opt for Extended warranty. Warranty covers the following support:

- 24 X 7 Telephonic, Email and Remote Service Support when required
- Regular Software updates and Bug Fixes.
- Supply of Mechanical and Electrical components in case of failure (excluding damages as mentioned in Exclusion Clause)

The warranty does not apply to replacement or repair of:

- Normal wear and tear.
- Consumables.
- Faulty articles continued:
 - Failure to comply with the manufacturer's recommendations (logistics documentation, Technical Information Note, retrofit document) and the rules of the trade.
 - Negligence or abnormal use of equipment.
 - Anomalies produced by an environment of use, storage or transport that does not comply with the specifications or recommendations of Falcon: packaging, temperature, hygrometry, sector, insulation, etc.
 - A defect due to a cause external to the supplies and services of Falcon.
- Equipment other than that supplied by Falcon.
- Items that can be repaired exclusively by Falcon that have been repaired or attempted repairs other than those carried out by Falcon.
- Items that fail due to normal wear and tear of one or more of its components or whose tamper-evident seals (varnish, strip, etc.) have been broken or whose serial numbers have been removed or modified.
- Items damaged during transport to Falcon due to the use of unsuitable packaging.



18. Exclusions

The scope of supply includes all parts which are defined in the Supplier's quotation.

All other parts which are not defined in the Supplier's quotation do not belong to the Supplier's scope of supply and are excluded. The following parts are also excluded:

- **Steel work is not specified.**
- **Server PC**
- **Roll cage trolley**
- Construction Power
- Building infrastructure; building structure, doors, fire exits, levelling devices, building extinguisher and fire alarm system, building heating and lighting system.
- Electrical power supply and wiring to the main control cabinets
- UPS for Controls and Drives
- Electrical Power for Installation
- Network Cabling up to the Main Server Rack
- Intermediate wiring to parts which are to be supplied by the Purchaser/others
- Emergency/Uninterruptable power supply
- Fire-alarm and fire protection devices
- Traffic and route markings
- Laydown Area
- Ram protection devices
- Cat walks, bridges, maintenance aisles and platforms- If not specified
- Assembly tools like forklifts and hoisting machines
- All kind of network incl. Local Area Network (LAN/WLAN), exceeding the scope described in Scope of Supply
- Any Kind of Civil work
- Any adjustment of the Supplier's scope of supply to local rules and regulations
- X-Ray machines
- Roller cages/ Pallets
- Simulation and 3D animation of the sorter system
- Interface with other equipment not specified in this offer.
- Provision of facilities for the control room (furniture, air conditioning, heating, etc.).
- The supply and installation of fencing around the different corridors.
- Any item specifically indicated as not forming part of the subject matter of the Seller's supply in the offer documentation.